

The possibility of utilizing the BOT system to reconstruct the war-damaged Public Projects in Sudan

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مستخلص:

تناولت الورقة البحثية إمكانية تطبيق نظام البناء والتشغيل ونقل الملكية (BOT) كآلية لتمويل مشاريع إعادة الإعمار في السودان بعد الحرب، وذلك في ظل التحديات الاقتصادية والقيود المفروضة على الموازنة العامة. وقد تمحور السؤال الرئيس للدراسة حول مدى قدرة هذا النظام على المساهمة الفعالة في إعادة بناء ما دمرته الحرب. وتهدف الورقة إلى استكشاف مدى كفاءة نظام BOT في دعم مشروعات البنية التحتية، وتوفير الموارد المالية والتكنولوجية، فضلاً عن بحث الجوانب القانونية والعقبات التشريعية التي قد تعيق تنفيذه في السياق السوداني. ونظراً لصعوبة الحصول على بيانات دقيقة حول حجم الخسائر، قام الباحثان بتصميم مصفوفة تقديرية استناداً إلى تقارير صادرة عن عدد من المؤسسات المحلية والدولية. اعتمدت الدراسة على المنهج الوصفي التحليلي، من خلال تحليل البيانات الثانوية والأدبيات ذات الصلة على المستويين الإقليمي والدولي. وقد خلصت النتائج إلى أن تبني نظام البناء والتشغيل ونقل الملكية (BOT) يمكن أن يساهم بشكل ملموس في إعادة الإعمار دون تحميل الموازنة العامة أعباء مالية كبيرة، مع تحقيق فوائد إضافية مثل نقل التكنولوجيا وتوفير فرص العمل. وأوصت الورقة بإعطاء الأولوية للمشروعات الحيوية والاستراتيجية، بما في ذلك الطرق والجسور والنقل ومحطات تحلية المياه ومحطات الطاقة، إضافة إلى المستشفيات والمراكز الصحية.

الكلمات المفتاحية: نظام البناء والتشغيل ونقل الملكية، إعادة بناء السودان، إعمار البنى التحتية.

Abstract:

The paper examines the possibility of applying the Build-Operate-Transfer (BOT) model as a financing mechanism for post-war reconstruction projects in Sudan, particularly in light of the country's economic challenges and constraints on the public budget. The question of this paper addresses the extent to which the BOT system can effectively contribute to rebuilding war-damaged infrastructure. Thus, the study aims to explore the efficiency of the BOT model in supporting infrastructure development, providing financial and technological resources. Moreover, identifying the legal and legislative obstacles that may hinder its implementation in the Sudanese context. Due to the difficulty of obtaining accurate data on the extent of the damages, the researchers designed an estimated matrix based on reports issued by various local and international institutions. The study employed a descriptive-analytical approach, relying on secondary data and relevant literature at both regional and international levels. The findings indicate that adopting the BOT model could significantly contribute to reconstruction efforts without placing a substantial burden on the national budget, while also enabling technology transfer and creating employment opportunities. The paper recommends prioritizing vital and strategic projects such as roads, bridges, water Plants, power stations, as well as hospitals and healthcare centers.

Key Words: Build-Operate-Transfer, War in Sudan, war-damaged infrastructure.

Introduction:

Economic development, socio-economic inequality, poverty, and infrastructure remain major challenges for developing and developed nations. As a result, several countries have sought appropriate means to finance projects to reduce inequality and enhance overall development (Sabri, Syafrinaldi, Yusri , & Surizki , 2024). During the 17th century, the United Kingdom adopted the Public-Private Partnership (PPP) model for public projects, which effectively addressed funding shortages and inefficiencies while improving the overall benefits of infrastructure development (Huijie , Lihong , Hang , & Yuanzhi , 2025). While in Vietnam, approximately 2,700 kilometers of public paved roads, representing 10% of the country's total infrastructure projects-are financed through Build-Operate-Transfer (BOT) schemes (Linh, Nguyen, Nguyen , & Vo , 2023). Similarly, the completion of China's Shenzhen Shajiao B power plant in 1984 under the BOT model marked a significant milestone in the country's adoption of the PPP framework for infrastructure development (Huijie , Lihong , Hang , & Yuanzhi , 2025).

The key motive for BOT adoption is an imperative need for capital and development (Linh, Nguyen, Nguyen , & Vo , 2023). Therefore, the BOT system has been applied in various Arab and African countries, including Sudan, Saudi Arabia, Egypt, Jordan, Nigeria, and Rwanda. Sudan had two experiences in this field: the Manara Water Project and the Highway Road between Atabara and Hia, 274.7 kilometers (Abdelrahman, Mohammad , & Hawari, 2018).

Public projects are essential for every country's economic growth. These projects, such as highways, railways, paved roads, ports, airports, bridges, tunnels, power plants, hospitals, municipal buildings, and other public-use facilities (Rajkumar, Anandakumar, Rajeshkumar, & Sampathkumar , 2024).

Building an essential infrastructure requires a long-term financial commitment. Innovative finance has been spurred by the vital demand for such projects, as well as recurring budgetary deficits faced by government agencies. Developing countries have a lack in national funds. Therefore, governments have asked the private sector to take part in long-term contractual arrangements for the funding, building, and operation of infrastructure projects (Rajkumar, Anandakumar, Rajeshkumar, & Sampathkumar , 2024).

Various governments often face obstacles in infrastructure development related to the lack of enough capital and technology. To fill the weaknesses of various countries in infrastructure development, there is the Build-Operate-Transfer Scheme, which is supported by third-party funds, namely the private sector. This model has gained popularity in several developing countries. For example, in Turkey, 98 out of 198 infrastructure projects, representing 50%, have been implemented using the BOT model (Sabri, Syafrinaldi, Yusri , & Surizki , 2024). India has also increased infrastructure investments in recent years, particularly to encourage greater private sector participation, given the constraints of existing infrastructure (Rajkumar, Anandakumar, Rajeshkumar, & Sampathkumar , 2024).

Sudan's economy has been severely affected by war at both macro and micro levels, impacting imports and exports, production, industries such as mining and construction, public and private services, and foreign financial transactions (Esraa Mahadi Ali Mohamed & Don-Eliseo, 2025). The country's Governments often face challenges in infrastructure development due to limited capital and technology. To address these gaps, the Build-Operate-Transfer (BOT) model has emerged as a viable solution supported by third-party financing from the private sector.

The economic growth dropped by 12.5% due to armed conflict (Esraa Mahadi Ali Mohamed & Don-Eliseo, 2025). At the microeconomic level, the conflict caused extensive damage to Sudan's infrastructure and industrial base, alongside significant losses to public and private properties. Numerous public-private institutions, including oil facilities, conglomerates, bank branches, and

telecommunications, located in major cities like Khartoum, Nyala, Madani, and Sennar, were looted, burned, destroyed, or forced to shut down (Policy & Programme, 2024). Sudan is now in urgent need of rebuilding and reconstructing public infrastructure projects that directly support economic growth and indirectly improve the socio-economic well-being of the population by creating job opportunities. These projects require substantial funding that the government currently cannot allocate. Therefore, the BOT system appears to be a suitable financing model, especially considering Sudan's limited access to financial and technological resources. The country has already experienced two BOT projects, including the successful Manara Water plant Project and the Atbara–Hia paved road (Abdelrahman, Mohammad , & Hawari, 2018).

Sudan has witnessed a sharp decline in its infrastructure due to ongoing conflicts, which have destroyed vital public assets. Given the government's constrained financial capacity, it is crucial to explore alternative mechanisms to finance and rebuild these projects. However, questions remain regarding the feasibility of implementing this model in Sudan's context, the willingness of the private sector to participate, and the legal and institutional challenges involved. Accordingly, the research problem is defined as follows: To what extent can the BOT model be utilized to rebuild war-damaged public projects in Sudan? The study aims to examine the effectiveness of the BOT model in financing and reconstructing such projects, assess the readiness of the private sector to participate in BOT-based initiatives, identify the legal and regulatory barriers to implementation, draw lessons from other countries' experiences, and offer practical policy recommendations tailored to Sudan's local context. The hypotheses of this paper suggest that adopting the BOT model in Sudan could enhance infrastructure development and improve the performance of these projects. Additionally, relying solely on public funding to finance such projects may increase the government's fiscal deficit. To address these issues, the study will adopt an analytical and descriptive approach, drawing on prior research and international case studies of BOT financing models.

Scientifically, this paper offers a solid literature basis on modern financing modes that have evolved from traditional franchise contracts. While, Practically, it demonstrates how the Sudanese Ministry of Finance can leverage the paper's findings and policy recommendations to implement strategic, high-impact projects. These initiatives aim to enhance citizens' quality of life while alleviating the financial strain on the public sector-an outcome that cannot be achieved without embracing innovative financing methods.

Data sources and Methodology:

The researchers will utilize secondary data and statistical information obtained from reports issued by the Central Bank of Sudan, as well as from international organizations such as the International Monetary Fund (IMF), the World Bank (WB), and the African Development Bank (ADB). Additional data will be sourced from peer-reviewed research papers and academic journals. based on these sources, the researchers will construct a matrix highlighting the most critically affected sectors in the Sudan due to the ongoing conflict, specifically, infrastructure including, the health sector, public and private education, and transportation. More precisely, the focus should be paid to water and power stations. This matrix will form the basis for identifying key findings and formulating evidence-based policy recommendations. Accordingly, the study adopted a descriptive and analytical approach, aiming to systematically document and describe the war's impact on these vital sectors, while also analyzing the broader implications and extracting insights to inform future policy recommendations.

Theoretical Literature Review:

3.1 Concepts and types:

The concept of the BOT differs according to different scholars and authors, as Khalid, Ran, & Ahmed F. M., (2014) defined it as “the granting of a concession by the Government to a private promoter, known as the concessionaire, who is responsible for financing, construction, operation and maintenance of a facility over the concession period before finally transferring the fully operational facility to the Government at no cost”. Also, Amusan, Joshua, and Oloke (2013) defined it as a process whereby the public sector can partner with the private one in executing capital projects and infrastructure. Similarly, this could be referred to as a form of project financing mechanism that helps a private institution obtain a concession from the public to finance, design, construct, and manage a facility.

As a legal term can be defined as a long-term contract settled between a governmental entity and a private investor, whereby the latter undertakes to build and operate a specific project for a set period, after which ownership of the project is transferred to the government (Al-Baghdadi, 2005).

3.2 Types of BOT Systems:

The Build-Operate-Transfer modes of finance have various sorts, one of which is BOT, and PPP, which stands for Public-Private Partnership, that can be understood by various interpretations. However, the general conception is that the state and private investors sign a joint contract, share the responsibilities, benefits, and risks to build infrastructure or provide public services (Linh, Nguyen, Nguyen , & Vo , 2023). Build-Operate-Transfer (BOT): This type of scheme has been used in power stations, drinking water supply, transportation, telecommunication, and process plant sectors (Mohammed & Babanyara, 2015).

Table (3.2.1) The type of BOT Model according to Operation

Type	Abbreviation	Who operate	Ownership During Operation	Description
BOT	Build-Operate-Transfer	Private Sector	Often public or shared	A private entity builds a project, operates it for specified period to recover its investment and then transfers ownership to the government
BOO	Build-Own-Operate	Private Sector	Private	Here, a private entity builds the project, owns it and operate it without government’s intervene.
BOOT	Build-Own-Operate-Transfer	Private sector	Private	A private entity builds the project, owns it and operate it, at the end transfers it to the authority
DBO	Design-Build-Operate	Private sector	Private	Here, the entity is responsible of designing the project, builds and entirely operate it.

DBOT	Design-Build-Operate-Transfer	Private sector	Private	Here, the entity is responsible of designing the project, builds and entirely operate it, transfers the project to the government
BLT	Build-Lease-Transfer	Public Sector through lease	Private	The private sector builds the projects, lease it to the public sector for specified period of time, then transfers the assets to the public authority at the end of the lease.
BTO	Build-Transfer-Operate	Private sector	Private	The private builds the infrastructure, then transfers ownership to the government immediately after construction

Source: (Abdelrahman, Mohammad , & Hawari, 2018)

The Sharia and Legal Framework of BOT System:

The BOT model is one of the modern mechanisms designed to engage the private sector in infrastructure projects without imposing additional financial costs or burdens on the state (Al-Tayeb, 2007). As reliance on this model increases in developing countries, questions arise regarding its compliance with Islamic Sharia and national laws, particularly in the Sudanese context (Mohammed A. , 2010).

Sharia Classification of the BOT System:

Though the BOT model is not explicitly mentioned in classical Islamic jurisprudence, it can be interpreted through the lens of several recognized Sharia contracts, such as:

a. **Ijarah Muntahia Bittamleek (Lease-to-Own Contract):** The project is leased to the investor for a defined period, after which ownership reverts to the state (Saeed, 2013)

b.**Istisna' Contract:** An agreement to construct a specific project based on predefined specifications (Abdullah, 2001).

c.**Musharakah (Partnership Contract):** The investor finances and operates the project, with ownership transferred at the end of the contract duration (Ishaq, 1997).

Sharia Evidence Supporting the Legitimacy of BOT:

The acceptability of the BOT system can be supported by various Sharia principles, including: The principle that "the default in contracts is acceptability," unless they contain prohibited elements (Ahmed Mohammed Al-Fiqh, 2010, p. 85).

The Prophet's Hadith (peace be upon him) "Muslims are bound by their conditions, except for a condition that makes the unlawful lawful or the lawful unlawful" (Al-Tirmidhi).

The legal maxim "*No harm shall be inflicted or reciprocated*": BOT projects serve the public interest and do not inherently involve harm or unjust exploitation (Mohammed A. , 2009)

Legal Framework of BOT in Other Jurisdictions:

Egyptian Law

In Egypt, the BOT system is governed by Law No. 67 of 2010 regulating private sector participation in infrastructure and public service projects (Executive Authority, *Official Gazette of Egypt*, 2010).

Jordanian Law

In Jordan, the BOT system is regulated under Public-Private Partnership Law No. 31 of 2015 (Executive Authority, *Official Gazette of Jordan*, 2015).

UAE Law

In the United Arab Emirates, specifically in Dubai, BOT is governed by Law No. 22 of 2015 on Public-Private Partnership (UAE Ministry of Economy, *Official Gazette of the UAE*, 2015).

3.3 Legal Framework of BOT in Sudan

The Investment Encouragement Act of 1999 and its Amendments

This is one of the core laws governing investments in Sudan (Ministry of Investment, *Official Gazette of Sudan*, 2003).

The Land Act of 1970 and its Amendments

This law regulates land ownership and use in Sudan (*Sudanese Land Act*, Khartoum, 2012).

The Companies Act of 1925 and its Amendments

It governs the establishment and operation of companies in Sudan (*Sudanese Companies Act*, Khartoum, 2011).

The Labor Act of 1997 and its Amendments

This law outlines the rights and obligations of workers in Sudan (*Sudanese Labor Act*, Khartoum, 2002).

The Environmental Protection Act of 2001 and its Amendments

It regulates environmental protection measures in Sudan (*Sudanese Environmental Act*, Khartoum, 2005).

(Legal Challenges in Implementing BOT in Sudan

Despite the economic benefits of BOT, there are legal challenges that must be addressed, such as the lack of a clear legal framework that effectively governs such contracts (Al-Baghdadi, 2005).

Risk associated with BOT:

There are various sorts of risks associated with BOT projects, including Socio-economic risk which encompasses fluctuating social service demand toll and change in stakeholder expectations. This often affects the psycho-social awareness of workers. Economic risks induced by the impact of the global economy, unstable exchange rates, which affect the fund flow on a project. Health & safety risk is a different type of risk that is health-condition induced. Noise pollution, emission of poisonous substances, vibration, biological hazards, and stress are regarded as health-induced risks. Also, BOT projects can be influenced by operational and technological risks. The first one includes building delivery and maintenance-related issues that constitute risks, while technology-based risk entails obsolescence of existing systems, cost of procuring the best technology, and implications. However, despite the range of associated risks, there is frequently a level of success incurred on a BOT project, which is determined by the success factors associated with BOT projects (Amusan , Joshua, & Oloke , 2013).

3.4 Empirical literature Review:

(Huijie , Lihong , Hang , & Yuanzhi , 2025) defined the term “Public-Private Partnership” as a sort of action that depends mainly on non-governmental organizations (NGOs) to meet people’s needs. The PPP model relies less on government action than formal projects and also points out that the key to success is competition among private institutions. (Sabri, Syafrinaldi, Yusri , & Surizki , 2024) have measured the development of research on national infrastructure development using the Build Operate Transfer (BOT) schemes. Their research has used a bibliometric analysis approach to disclose the mapping of studies and research trends regarding BOT. Another study by (Rajkumar, Anandakumar, Rajeshkumar, & Sampathkumar , 2024) explored the risks, delays, success factors, contractor selection, and best value selection models in construction project management. Their

study has assessed the extent of time overruns in building projects by conducting a postal survey using a three-part questionnaire.

(Huijie , Lihong , Hang , & Yuanzhi , 2025) used Cite Space software to analyze the sample authors, journals, and regions in the Scopus database. Their paper aims to find out the recent development status, research paradigms, and research gap, as well as future trends of the PPP model. Their findings were that the concentration on PPP research has shifted from traditional models such as Build-Operate-Transfer (BOT) and Private Finance Initiatives (PFI) to contemporary themes such as risk management, policy analysis, and project governance.

(Tariq, 2024) studied the BOT system in Islamic Sharia, h .His research is based on three methods, one of which is an inductive, deductive, and analytical approach. He found that all types of BOT systems do not contradict Islamic Sharia since they represent a legal contract between the government and the private sector to build a national project. He added that there are some Quranic verses and Hadiths proving the applicability of this type of finance.

(Abdelrahman, Mohammad , & Hawari, 2018) Addressed the public-private partnership in infrastructure projects, and focused on the BOT system, which is broadly used in the partnership between the public and private sectors. BOT is considered a vital system in the economies of both developed and developing countries. Some of the leading Arab experiences in the use of the BOT system, the study concluded that the BOT technology is effective, and has great economic, social, and environmental importance.

(Linh, Nguyen, Nguyen , & Vo , 2023) have used the collected data from Vietnam, where over 2000 kilometers of public roads funded by BOT modes, they employed a quantitative methodology to measure the impact of some factors through a derived multivariate regression equation. Their findings disclose the existence of five separate groups of factors that affect the investment attractiveness of BOT projects in the Southern Key Economic Zone, classified by their levels of influence.

(Ali & IbnOuf, 2024) studied the BOT system implemented in public projects in Sudan, with a main focus on two public projects: Manara Water and another one is the Highway Road between Atbara and Hia in the northeast. One of their main findings is that the Water Corporation had achieved short-term objectives, including establishing the project, and the production and distribution of water. Nonetheless, the investors' rights and financiers, as well as paying back their return on capital, have been achieved, which may raise concerns that this project may have negative effects on other future ones.

Jiaju, Tahir, & Guru (2017) aimed to create a list of Critical Success Factors for China's BOT projects to support organizations in achieving project success. In their study, a questionnaire survey was conducted to find the relatively important CSFs for BOT projects in China. their initial assessment suggests that the key CSFs correspond to our defined categories of BOT projects' success factors. They found out that the list of CSFs could play a useful role in the phases of preparing, bidding, constructing, operating, and transferring of BOT projects.

The study (Mohammed & Babanyara, 2015) compared common practices of BOT-type projects in different countries and industrial sectors. They analyzed experience lessons by studying examples of both good practices and failures. Their findings show that BOT schemes are suitable for developing countries to construct infrastructure projects.

Amusan, Joshua, & Oloke (2013) focused on the cost implications of risk factors on the performance of Build Operate Transfer (BOT) projects. A questionnaire of 170 questions was used and analyzed by content analysis. They found out that the most common risks to the projects executed are inflation, variation to works, change in government policy, and fluctuating nature of foreign exchange, with inflation being the highest on the rating scale of 0.1 to 1.0, with corresponding cost implications and years.

Llanto (2008) has pointed out the main issues and constraints facing the successful implementation of the BOT model to infrastructure provision in the Philippines. He also indicated numerous factors that were contributory to an effective public-private partnership in BOT projects. Additionally, He pointed out the need to address various issues, starting from the legal basis to the responsibilities of the government entities that are involved in the project cycle, for instance, from project entry level to implementation and completion. Improvements should be introduced at the policy, legal, and institutional frameworks to enhance the usefulness of this approach to infrastructure development. Finally, he recommended that the government should provide an enabling framework for private-public partnership in infrastructure provision, which allocates roles, functions, powers, duties, and rights of the government and the private sector.

Table (3.2.2) Projects financed by BOT models in the Middle East (in million USD)

Country	Project's name	Financing	The project's cost	Contract Model	Time of Transfer
Saudi Arabia	Rabigh 3 desalination plant	Limited recourse project financing was arranged by Natixis, MUFG, Samba, and Riyadh Bank; Riyadh Bank provided an equity bridge loan	\$ 687 million	Build-Own-Operate (BOO)	25 years
Jordan	Samra Power Plant	Samra Electric Power Company	\$ 857 million	BOT	NA
Egypt	North Giza Power Plant	World Bank, European Investment Bank, and OPEC	WB \$600 million EIB €50 million OPEC \$30 million	Build-Operate-Transfer (BOT)	NA
United Arab Emirates (UAE)	Mohammed bin Rashid Al Maktoum Solar Park (Dubai)	Abu Dhabi Power Company and other companies	\$ 13.6 billion	BOT	NA

Source: ACWA Power.com

Table (3.2.3), Infrastructure projects in Africa financed by BOT system (in million USD)

Country	Project's name	Financing	The project's cost	Contract Model	Time of Transfer
Sudan	Al Manara Water Treatment Plant	The Dutch government and Biwater company	106 million (50% covered by the Dutch government) and 50% by Biwater (a British	Build-Operate-Transfer (BOT)	15 years (starts 2010 and ends 2025)

			company)		
Rwanda	Kigali Bulk Water Supply Project	Metito (UAE-based)	\$60 million	Build-Operate-Transfer (BOT)	27-year contract
Nigeria	Gas-fired power plant (Azura-Edo Independent Power Project)	International finance institutions like IFC, OPIC, and several European DFIs	\$900 million	Build-Operate-Transfer (BOT)	-15year contract

Source: (Bank, 2015), and (Jackson, 2015)

3.4.1 Implementation of Public Infrastructure Projects in Africa and Asia:

Al Manara Water Treatment Plant costs about 106 USD million, 50% financed by the government of the Netherlands and 50% by the Biwater British company. Al Manara is located in Omdurman, Khartoum State, Sudan. The main purpose was to enhance access to potable drinking water in the rapidly growing urban area of Omdurman with a capacity of approximately 180,000 m³ per day, serving around 1.2 to 1.4 million residents. The Operational Period is a 10-year operation phase following a 3-year construction period. Then, followed by the Transfer -the Ownership and management to be transferred to Khartoum State Water Corporation (KSWC) after loan repayment (Jackson, 2015). The project was fully financed without a debt burden on the Government. It was the first water project in Sudan to be funded completely by the private sector. The financing partners involved the Netherlands Development Finance Company (FMO), the Netherlands Government-supported ORET Grant, Export Credit Insurance Corporation of South Africa Limited (ECIC), Industrial Development Corporation of South Africa Limited (IDC), Export-Import (EXIM) Bank of Malaysia, and Calyon Bank of France (Jackson, 2015).

Similarly, the Kigali Bulk Water Supply Project was a successful example of the partnership between the public and private sector called: the private Partnership (PPP) between the Government of Rwanda (GOR), represented by the Ministry of Infrastructure, Water, Sanitation Corporation (WASAC), the government water utility company and Kigali Water Limited (KWL). The project scope involves the development, design, financing, construction, and operation of a 40,000 m³ per day Bulk Water Facility (BWF) south of Kigali in Rwanda. The scope of the BWF comprised of; a water treatment plant, a well field with 38 wells, three pumping stations, pipelines, and three storage reservoirs. Under the PPP arrangement, Kigali Water Limited will supply bulk water to the Water and Sanitation Corporation, which WASAC then sells it to the local consumers (Bank, 2015).

This project is expected to have solid development outcomes firstly by providing hygienic potable water to the population contributing to improving public health, secondly, addressing Rwanda's growing demand in the industry sector for reliable water, thirdly causing economic growth due to improved productivity, particularly for women and redeployment of Government budgetary savings to the health sector and other economic sectors. Finally, the strong demonstration effect given that this is the country's first Independent Water Producer. project on a large scale in the continent, therefore contributing to private sector development (Bank, 2015).

As for Nigeria's Azura-Edo Independent Power Project, the major expected infrastructure development benefits of the project entail delivering much-needed power to Nigeria, and, in turn, the broader West African power grid. In doing so, the project is expected to give access to affordable electricity to about 14 million residents at the cost of self-generated power. The Azura-Edo IPP also has a strong demonstration effect and sets a model for future private sector investors in the gas-to-power value chain. It has high levels of government support, a devoted lead sponsor,

robust technical and engineering support, and was situated both near the country's main gas trunk line and the national transmission grid (Development , 2017). This project includes as project sponsors a consortium of private investors (97.5 % and the Edo State government 2.5%. The consortium is composed of a joint scheme between Amaya Capital Ltd. and Actis GP LLP (51 percent), AIIF2 Power Holding Ltd. (29 percent), Aldwych Azura Ltd. 14%, and ARM- Harith Infrastructure 6%. The plant will have a sole off-taker, the state-owned Nigeria Bulk Electricity Trader (NBET) under a 20-year power purchase agreement (PPA) backstopped by a Put Call Option Agreement (PCOA) with the Federal Government of Nigeria. Seplat Petroleum Development Company, a Nigerian upstream production and development company, will supply gas to fuel the plant under a 15-year contract.

In terms of the Middle East Saudi Arabia's desalination plant proposed by the Consortium has been designed to comply with the water output supplies and to guarantee the required availability of potable water. The design of the plant and the related buildings have been developed focusing on the combination of operational suitability, plant availability, optimization of power consumption, energy efficiency, and minimization of the environmental and visual impact. The project was financed by a limited-recourse project by Natixis, MUFG, Samba, and Riyadh Bank; Riyadh Bank provided an equity bridge loan. It costs approximately \$ 687 million paid after 25 years through the BOT system (ACWA Power, 2019).

Finally, Mohammed bin Rashid Al Maktoum Solar Park (Dubai) follows a public-private partnership (PPP) model, including BOT structures, with the involvement of global companies such as ACWA Power. It has a planned production capacity of 5,000 MW by 2030, with investments totaling AED 50 billion. When completed, it will save over 6.5 million tons of carbon emissions annually.

The total capacity of the sixth Phase is 1800 MW using photovoltaic solar panels, based on the Independent Power Project (IPP) model. The sixth Phase will reduce around 2.36 million tons of carbon emissions annually and provide clean energy to 540,000 residences in Dubai. Through phase VI, DEWA has achieved the lowest Levelised Cost of Energy (LCOE) of USD1.6215 cents per kilowatt hour (kWh). The 6th phase will become operational in stages between 2024 and 2026. (Sustainability & Innovation Centre, 2024).

(3.4.2) Sudan's Macro and Microeconomic Indicators during the War:

The Sudanese government planned to increase its revenue to 12.4% of Gross Domestic Product (GDP) in comparison to 9.7% in 2022. Public expenditure was planned to surge slightly from 11.2 per cent in 2022 to 13.8% in 2023, thus maintaining a budget deficit of only 1.4% of GDP to support the reduced inflation to 25% and achieve economic stabilization. The ongoing conflict will have a great effect on the fiscal policy and, accordingly, on the stabilization goal of the budget (Siddig, 2024).The armed conflict which erupted in Sudan between the Sudanese Armed Forces (SAF) and the Rapid Support Forces (RSF), created a multifaceted socio-economic and political situation. This conflict started in Khartoum but swiftly extended to the Darfur and Kordofan regions before impacting every state in the nation to varying degrees. In August 2024, the conflict had caused over 9,323 incidents and 20,013 fatalities. 50.6% of the incidents occurred in Khartoum State, the largest urban area in Sudan, followed by 11.07% in Al Jazirah and 8.04 % in North Darfur State. After one year, 10.5 million Sudanese are estimated to have been displaced, of which 2.4 million have fled to neighboring countries such as Libya, Egypt, Chad, and Ethiopia. It is the worst displacement crisis in the world. Prior to the conflict, Sudan already had an estimated 2.8 million IDPs (IMO, 2024).

Previously, Sudan's socioeconomic indicators displayed a low development level as compared to those of its peer countries. In 2022, 19% of the Sudanese were below the \$ 2. 15 poverty line,

compared to Ethiopia (17%), Benin (12 %) and Senegal (10%). In 2022, Sudan ranked 170 out of 193 countries on the Human Development Index (HDI), one of the lowest globally (Research & Programme, 2024).

The macro level displays that the conflict has caused a significant worsening in economic activity. The Sudanese economy contracted by 20% in 2023 and 15% in 2024, while the African Development Bank estimates a GDP reduction of about 5.9% in 2024 and estimates an economic output loss of USD 125 billion in the first year of the conflict. Other estimates suggested that Sudan's GDP could have dropped by over 40 percent in 2023. Reinforcing these assessments, Sudan's Minister of Finance has reported in 2023 that the war had caused an economic loss exceeding USD 26 billion, or more than half of the nation's total GDP in 2022 (Khalid Siddig, 2025). Also, the inflation rate increased from 67% in 2023 to 137% in the first half of 2024. The World Bank estimates showed that Sudan's inflation rate is 66% in 2023 and 180% in 2024, while the estimates of the International Monetary Fund are 171% and 145% in the same periods. As for August 2024, the Sudanese currency has depreciated by over 300% since the start of the conflict in April 2023. The number of poor people in Sudan has increased by 1.8 million in 2023 (Research & Programme, 2024)

At the micro level, the ongoing conflict has hit Sudan's economic centers hard, causing huge damage to the country's infrastructure and its industrial base, as reports showed that over 600 factories are out of service, as well as huge losses to both public and private properties. All economic public-private entities of all sizes and structures, including oil industry facilities, large conglomerates, bank branches, and telecommunication, located in urban centers like Khartoum, Nyala, Wad Madani, and Sennar, were looted, burned, destroyed, or forced to close (Research & Programme, 2024).

The ongoing conflict has negatively impacted the delivery of basic services in Sudan. 88 attacks on healthcare facilities. In addition to disease outbreaks, there are severe shortages of medical supplies and liquidity to manage healthcare centers and pay salaries (World Health Organization, 2024). Less than one-quarter of health facilities are working in the Sudanese states hardest hit by the war, and only 45% of these facilities are entirely working in other states. Currently, about 15 million people need urgent health support for their survival, and an estimated 3.4 million children under 5 years old are at high risk of epidemic diseases.

As for general education, all Sudanese basic and secondary schools have been closed. At least 2,500 schools, 13% of all schools, were being used as Internally Displaced Persons (IDP) shelters, affecting the education of around 2 million children in the host communities. Many school buildings have been severely damaged, and rehabilitation is needed. Substantial damage to water and sanitation systems, electricity and energy infrastructure, paved roads, and the destruction of the transportation system. The public services provision in most areas has been badly impacted (UNICEF, 2024).

Table (3.4.2.1), number of public-private universities and colleges in Sudan

State	Number of public-private universities and colleges	Percentage
Khartoum	68	54
Gezira	15	12
Blue Nile	2	2
Sennar (Sennar)	1	0.8
Kordofan	3	2.4
Darfour	8	6.4

Source: (Beshir , Ahmed , & Mohamed, 2020)

The table above shows 12 public universities and 16 private universities in Khartoum State, in addition to 1 technical college and 39 private colleges. While Gezira State has 3 public universities, 1 private university, 4 technical colleges, and 7 private colleges, it represents 12% of the total number of Sudanese universities. As for Kordofan, Darfur, and Sennar have 12 universities, including public and private ones. Also, the table notes that Khartoum State represents 54% of the total number of 125 public and private universities, as well as private and technical colleges (Beshir , Ahmed , & Mohamed, 2020). The recent research paper indicates that the infrastructure of most of these universities was destroyed as a result of the ongoing war in Sudan, which affected more than 3 million students who were studying in these universities.

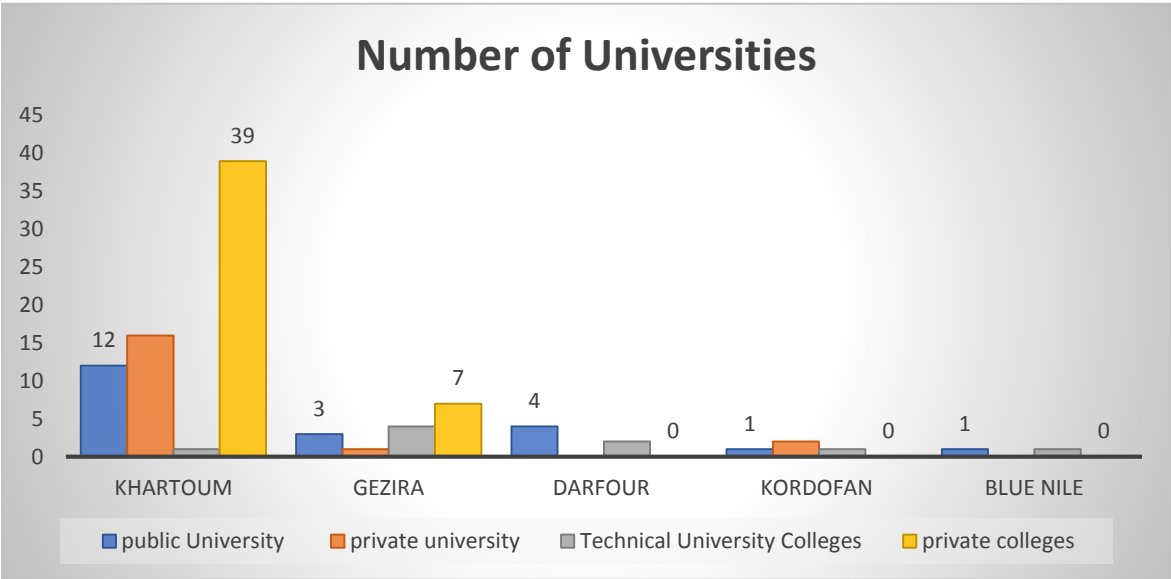


Figure 3.4.2.1The number of Universities in war areas in Sudan Source: Central Bank of Sudan, 2019-2020.

Table 3.4.2.2 Number of public-private hospitals in 2020 in Sudan

State	Hospitals	Health centers	Health Units	Total
Khartoum	53	451	132	636
Gezira	93	402	478	973
Blue Nile	18	68	72	158
Sennar state	34	124	254	412
Kordofan	70	329	614	1,013
Darfour	70	360	464	894

Source: Central Bank of Sudan (CBOS) 2019-2020

This data is derived from the Central Bank of Sudan's 2020 report. Therefore, the data represents approximate figures for the situation in Sudan during the war period. From this data, a general concept and matrix can be constructed regarding the damage to the health sector in Sudan as a result of the 15th April 2023 war. In this table, we have summarized the number of states in Sudan, focusing on the states most affected by the war. The table above indicates the number of hospitals and health centers that have become out of service, leading to a deterioration in the health situation in war zones. This has led to a health disaster in all states of Sudan, negatively impacting citizens'

health. The numbers above show that there are 4,086 hospitals, health centers, and health units that are out of service.

Table 3.4.2.3 Estimated Matrix of Sudan’s Public-Private Losses and Damages

Sector	Losses and damages	Estimated Values	Estimating Authority
Health	,0864	NA	Wikipedia
High Education	97 universities	50 billion Dollar	Ministry of Higher Education
General Education	Over 70% of the schools closed causing 7,544,024 students stopped learning due to the war	NA	Ministry of General Education
Banks and Financial Institutions	38 governmental and commercial banks were completely damaged. In addition to 6 foreign banks stopped working.	NA	Central Bank of Sudan
Infrastructure	Bridges, Transport system, Telecommunication and power stations	1 billion dollars	Ministry of Transport
Agriculture and Natural Resources	Agricultural schemes were out of production	NA	Ministry of Agriculture and animal wealth
Factories and SMEs	Over 600 factories stopped working. In addition to SMEs.	300 billion dollars	Sudanese Business Federation

Source: The authors based on Annual Reports of the Central Bank of Sudan and other sources included in the above table.

4. Findings and Conclusion:

The ongoing conflict in Sudan, which began in April 2023 between the Sudanese Armed Forces (SAF) and the Rapid Support Forces (RSF), has had a catastrophic impact on the country’s economy, infrastructure, and public services. Despite initial government plans to improve fiscal performance and economic stability in 2023, the war caused Sudan's economy to shrink by 20% in 2023 and 15% in 2024, with inflation soaring to over 130% and the currency depreciating by more than 300%. The conflict has displaced 10.5 million people, making it the world’s worst displacement crisis. Basic services, including healthcare and education, have been severely disrupted. Only a fraction of health facilities remains operational, and schools across the country are closed, with many repurposed as shelters. Infrastructure damage has paralyzed industrial and service sectors, particularly in Khartoum and other urban centers, affecting over 600 factories and numerous economic institutions. Higher education has also suffered, with the majority of the 125 universities and colleges damaged, disrupting the education of more than 3 million students. With over 20,000 deaths and widespread destruction, the war has deepened poverty, reversed development gains, and severely undermined Sudan’s prospects for economic stabilization and recovery.

The reviewed literature examined the experiences and case studies of several countries, including China, Vietnam, Egypt, Saudi Arabia, Jordan, Rwanda, and Nigeria. The research findings from these countries consistently affirm the effectiveness of this modern form of financing as a practical and innovative alternative to traditional mechanisms, such as the franchise system previously granted to financing entities.

The paper findings align closely with those presented in the literature such as (Abdelrahman, Mohammad , & Hawari, 2018), (Al-Tayeb, 2007), (Ali & IbnOuf, 2024), and (Amusan , Joshua, & Oloke , 2013). They emphasize the importance of maintaining transparency in the selection of financing agencies for various projects, in order to mitigate potential drawbacks associated with opaque processes. Moreover, the study underscores the necessity of implementing robust oversight mechanisms to monitor the performance of these agencies. This is particularly crucial to ensure the eventual transfer of project ownership to local authorities, in accordance with international best practices and standards.

To answer the questions and hypotheses posed in the paper, the researchers, through a review of the relevant literature and recent reports on Sudan related to the paper's topic, have found that:

1. The Sudanese state is facing a severe shortage of domestic financial resources, exceeding USD 150 billion, as well as a lack of adequate technology needed to rebuild the war-damaged public projects in Sudan.
2. There has been destruction and loss of infrastructure in all cities affected by the war, such as Khartoum, Al Jazirah, Sennar, and western Sudan. The impact even extended to safe areas, where power and water stations were destroyed.
3. The country has lost most of its export cash crops and livestock, including sheep, camels, cattle, Gum Arabic, and soybeans. This is due to the Rapid Support Forces' control over production areas such as Kordofan and Darfur, as well as Al Jazirah and Sennar states.
4. The infrastructure of the financial and monetary sector-including banks, ATMs, and financial markets, has been destroyed. As a result, clients are now only able to deposit and withdraw money in limited areas, leading to financial exploitation and the spread of the parallel market. Consequently, public trust in the banking and financial sector has deteriorated.
5. The telecommunications infrastructure, particularly that based in Khartoum State, has been destroyed. This has led to widespread communication blackouts across large geographical areas of Sudan.
6. Factories operating in most Sudanese states, especially those affected by the war, have been destroyed. Over 600 factories ceased operations, leaving thousands of workers in both the industrial and agricultural sectors unemployed.
7. Health service centers across the country have collapsed or gone out of service, resulting in the spread of diseases and epidemics such as dengue fever, cholera, malaria, and others.
8. Since the onset of the war, higher education, comprising universities and higher institutes, as well as general education, has come to a halt in Sudan. Consequently, a large number of students have been deprived of continuing their education.
9. Based on available data, reports, and previous literature, the paper concludes that the BOT (Build-Operate-Transfer) system can serve as an alternative and effective mechanism for rebuilding what the war has destroyed in Sudan, due to its reliance on both local and foreign investments.
10. The paper also concludes that reliance on the BOT system would help provide substantial financial resources for the state, offer access to advanced technology, and create job opportunities amid the country's economic stagnation.

5. Policy Recommendations:

Regarding the critical challenges in Sudan's post-conflict reconstruction, and based on the analysis of existing data and literature, the following recommendations are proposed to effectively leverage the BOT system as a strategic mechanism for national recovery:

1. The government should Prioritize Strategic Sectors for BOT Implementation by focusing on the most critical and high-impact sectors, such as the infrastructure, including rebuilding the roads, bridges, power stations, and water treatment facilities in key cities (e.g., Khartoum, Al Jazirah, Darfur). The same is true for Health and Education. Rebuild hospitals, clinics, schools, and universities to resume public services.
2. The government should create a BOT Legal and Regulatory and transparent Framework, investor-friendly legislation to govern BOT agreements, as well as forming a national BOT authority to run and regulate project proposals, contracts, timelines, and compliance by including clear terms for risk-sharing, dispute resolution, and eventual asset transfer to the country.
3. Also, the government should take serious steps towards attracting local and foreign investors by offering tax incentives, guarantees, and land grants to attract their participation. Moreover, working with international financial institutions to obtain credit guarantees or co-financing schemes. Promote BOT opportunities through international donor conferences and investment forums.
4. On top of that, more steps must be taken to enhance Public-Private Partnership (PPP) Capacity by creating PPP units within main ministries, for instance, infrastructure, health, and education. Additionally, qualify local staff in negotiation, risk management, and financial modeling for BOT projects, and encourage partnerships between Sudanese firms and international contractors.
5. As for the financial sector infrastructure in the war-damaged areas, the government should restore the financial sector by adopting BOT models for rebuilding banking networks, particularly digital financial infrastructure and rural access. And collaborate it with fintech firms to provide mobile banking solutions in underserved areas.
6. To guarantee that the BOT projects are performing well, the government must control and evaluate BOT Projects by developing a performance monitoring system to ensure efficiency, transparency, and public accountability, as well as civil society organizations, and local communities should be involved in this process.

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